A Rescue That Knows Its Group: Training Deadlocks and Partial-Credit Traps in Learning to Compose from Pixels

Sankalp Pathak[1], **Sanjay Garg**[1], **Naveen Saini**[2]

[1] Jaypee University of Engineering and Technology, Guna, Madhya Pradesh, India

[2] Indian Institute of Information Technology Allahabad, Prayagraj, Uttar Pradesh, India

Abstract

Small vision transformers asked to compose elements of a finite group from images often fail to learn at all. We study why on a pixel-native benchmark in which the same group can act either on where glyphs sit or on what colour they are. First, a failing run is stuck in a training deadlock, not limited in what it can represent. A probe with a positive control finds the composite absent from a failing model rather than present and unreadable, and a small model trained first on a one-step version of the task breaks the deadlock in every seed when it knows the same task, in nearly every seed when it knows the same group acting on the other kind of picture, and no more often than a cold start when it knows a different group. That specificity held twice on fresh seeds whose tests were written down before they were run. Second, a visible cue that is a true partial answer to the task, one that composes, traps training: runs that pick it up stop at the score that cue alone earns, a prediction that also held on fresh seeds registered in advance. Third, how far apart a rendering draws the group's elements, computed from the renderer and fixed before it was measured, tracks how often seven renderings are learned. The models learn the group far more often from position than from colour at the setting we first used, but that gap was not shown to survive a change of learning rate and weight decay or of tile scale, and we report it as a setting-specific observation. Looped and feedforward models fail at indistinguishable rates, so none of this is about recurrence.

1. Introduction

A network that has to compose two group elements shown as pictures has to read each picture, bind what it reads to the right operator, and combine the two in order. We built a benchmark in which the number of compositions is known exactly and the same group can act on two different things: where glyphs sit, or what colour they are, with the group's order, generators and order-blind ceiling held fixed. We set out to test whether depth recurrence helps with that kind of binding, and at matched compute it does not. What we found instead concerns how such models fail to learn, and what gets them out.

Most runs that fail do so the same way. They stay at chance, emitting one class, for their whole budget. This paper is about that failure and makes three contributions.

1. **A training deadlock with a rescue that knows its group:** a failing run does not compute the composite at all; a probe that reads it in a solved model, at the same sites, reads nothing in a failing one. A donor, a model trained first on a one-step version of the same group, breaks the deadlock within a few thousand steps. The rescue is specific to the group and transfers across substrates: a donor of the same group acting on the other kind of picture works, and a donor of a different group does not. Both halves held on two sets of fresh seeds whose tests were fixed before the runs existed. Section 5.3.
2. **A true partial answer that traps training:** adding to the colour rendering a visible cue that carries one composing bit of the answer does not help. Runs that pick it up stop at the score that bit alone earns, and that held as a registered prediction on fresh seeds. Unlike the spurious features studied in shortcut learning (Geirhos et al., 2020), this cue is a correct quotient of the label, so the trap is partial credit on the real task. Section 5.4.
3. **Salience, measured before it was correlated, tracks learnability.** How far apart a rendering draws the group's elements, in pixels, was defined, implemented and tested on synthetic tiles before it was computed, and its rank correlation with escape rate across seven renderings cleared the threshold fixed in advance. Seven points make it suggestive rather than significant. Section 5.5.

The asymmetry that motivated the study, that the group is learned far more often from position than from colour, is real at the setting we first used and held there on fresh seeds. It was not shown to survive a change of learning rate and weight decay, or glyphs spread over several patches, and Section 5.2 reports it with those limits. Every test in this paper was registered before its runs existed unless it is labelled post hoc.

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2. Related work

Universal transformers (Dehghani et al., 2018) and recurrent-depth language models (Geiping et al., 2025) apply a shared block repeatedly, and recurrent vision networks that iterate longer at test time extrapolate on mazes and prefix sums (Schwarzschild et al., 2021; Bansal et al., 2022). None of those tests whether recurrence helps binding specifically, and we found no prior work that does. Step-norm and KL exit criteria come from Geiping et al. (2025); Pappone et al. (2025) propose a second-order exit rule that they report beats a KL exit and a first-order variant; Popescu et al. (2026) find simple confidence readouts often match or outperform learned halting gates. Appendix B scores such criteria against an oracle the benchmark provides.

Greff et al. (2020) set out the binding problem for artificial networks. Huang et al. (2026) formalise it information-theoretically and measure binding information in pretrained vision transformers; their

abstract does not describe manipulating spatial and appearance features independently on a fixed group, which is what our benchmark does.

Shortcut learning (Geirhos et al., 2020) and simplicity bias (Shah et al., 2020) describe models that latch onto easy, spurious features. Hermann and Lampinen (2020) show that an easier, weakly predictive feature can suppress a more strongly predictive one, and that a model's preference tracks what is most linearly decodable from the untrained model; Hermann et al. (2023) separate a feature's predictivity from its availability. The trap of Section 5.4 is an instance of an available feature winning, and differs in what the feature is: it is not spurious but a true quotient of the label, so the model that takes it is right about part of the answer and then stops. Our salience result in Section 5.5 is in the spirit of an availability measure, computed from the renderer rather than from a network.

Networks trained on finite group operations learn representation-theoretic circuits whose identity and order of emergence vary across runs (Chughtai et al., 2023), and shallow transformers can compose group elements through shortcuts (Liu et al., 2022). Delayed generalisation on small algorithmic tasks is the grokking literature (Power et al., 2022; Nanda et al., 2023); our benchmark's validity gate saw it directly, in a first training budget that stopped before the task was learned.

Control tasks (Hewitt and Liang, 2019) measure a probe's selectivity by giving it random targets. Our control asks a different question: whether the same probe, at the same sites, finds the information in a model known to contain it. Only that second control lets a null probe on a failing model count as evidence that the information is absent. The coda lens we use is Lu et al.'s (2025), and our appendix result with it replicates theirs in a different modality.

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3. The benchmark

Three families share one vocabulary and one renderer.

- **Family A, depth:** a composition of group elements applied to a scene. Depth is the number of operators composed. The answer is the ordered composite.
- **Family B, depth with breadth held fixed:** the k-min curve family.
- **Family C, breadth:** breadth 4 to 8 at depth 1, so parallel work varies while sequential work does not.

The same group can act on either of two substrates. In the *position* condition the group moves where a glyph sits and how it is oriented. In the *appearance* condition, written `identity` in the code, the same group permutes the glyph's colours. Order, generators, abelianisation and the analytic order-blind ceiling are identical between the two. The only thing that differs is what the group acts on, and at the level of pixels that is a lookup table of eight (D4) or six (S3) small rendered tiles: in the position condition each element has its own silhouette, and in the appearance condition every element has the same silhouette and differs only in where its colours sit. Every glyph fills one 4 by 4 patch of a 32 pixel canvas.

We use D4, of order 8, and S3, of order 6. D4 is generated by a quarter turn and a reflection about the main diagonal. At depth 2 the order-blind ceiling, the best accuracy obtainable by a model that knows which operators appeared but not in what order, is 0.8125 for D4 and 0.7500 for S3, computed exactly over all operator pairs before any run. A score above the ceiling is evidence of ordered composition and not of set membership.

Figure 1 shows the benchmark itself, in real samples from the generator. Family A raises the operator count from 1 to 3 at a fixed breadth of 4, family C raises the object count from 4 to 8 at a fixed depth of 1, and the third row puts the two substrates side by side at matched depth and breadth.

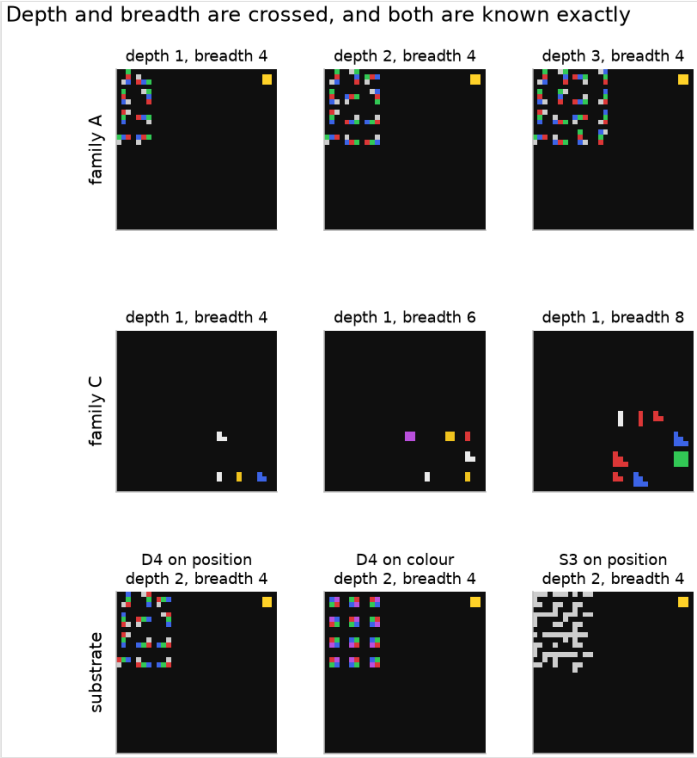


Figure 1. Real samples from the generator at a fixed seed, each labelled with the depth and breadth read off its own program. Top, family A at depths 1 to 3 and breadth 4. Middle, family C at breadths 4 to 8 and depth 1. Bottom, the same group acting on position and on colour at matched depth and breadth; this row needs colour to read.

4. Models and training

Every model is a vision transformer built from a prelude, a core and a coda. The looped model applies the core k times; two feedforward baselines stack distinct core blocks, one matched to the looped model's compute and one to its parameter count. All share the prelude, coda, output norm and head, so the arms differ in weight sharing and depth rather than in interface.

Unless stated, models have width 384 and 2, 2 and 2 blocks in prelude, core and coda, about 11M parameters for the looped model, and train for 300000 steps at learning rate 3e-4 and weight decay 0.05. Runs added for the robustness and confirmation tests of Sections 5.2 to 5.4 stop early once validation

accuracy has stayed at or above 0.95 for 20000 steps, a rule fixed before those runs and checked on earlier runs, none of which finished below 0.95 after reaching it.

A run escapes if it finishes at or above 0.95. Nearly every run finishes either there or on a floor, so escape is the unit the rates in this paper count, and no failed run is left out of them.

A perfect score gets three controls: a blank-image control, an input ablation, and the analytic ceiling above. The solved D4 model needs its state and both operators, correctly ignores distractors, and its 1.0000 sits above a computed ceiling of 0.8125.

The benchmark was validated before anything was measured on it, and passed on its second attempt. A depth 3 task was unsolvable at one pass (0.0212 against a 0.0208 chance floor), a depth 1 task was solvable at one pass, the order-blind ceiling sat below the threshold at every depth, and family C was solvable at every breadth. The first attempt failed only because its budget stopped short: the task was learned abruptly at about step 84000, the delayed generalisation known as grokking (Power et al., 2022).

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5. Results

5.1 What a failing run looks like

Of 196 finished family A runs at the time we counted, 104 finished above 0.95, 91 sat on a floor, and one landed between. Of the finished depth 2 failures that recorded it, 72 of 78 emitted exactly one class for all 16384 evaluation items, so the floor accuracy is the frequency of that constant class and not a measurement of anything the model computes. Escape is close to all or nothing, which is why we report rates of it.

5.2 The motivating asymmetry, and its limits

At the setting described in Section 4, the same group is learned far more often when it acts on position than when it acts on colour. Three competing predictions were registered before either new cell was built.

The 2x2 is complete: 70 of 70 seeds finished. Every cell carries at least the ten seeds the preregistration requires of a number that reaches a headline.

	acts on position	**acts on identity**
D4, order 8	9 of 10	12 of 40, floor 0.1210 to 0.1294
S3, order 6	6 of 10, floor 0.1681	3 of 10, floor 0.1681 to 0.1787

Identity is 15 of 50 finished across both groups and position is 15 of 20.

A two-sided Fisher test gives p = 0.0011. At this setting the escape rate follows the substrate and not the group: the registered alternative, in which the group's abelian part sets the rate, predicted the opposite in

both new cells and is refuted, as is the interaction alternative. **Figure 2** plots one point per seed rather than a mean, because the cells are not uniform. Fresh seeds held out of this grid, five for each of three cells, do not contradict it.

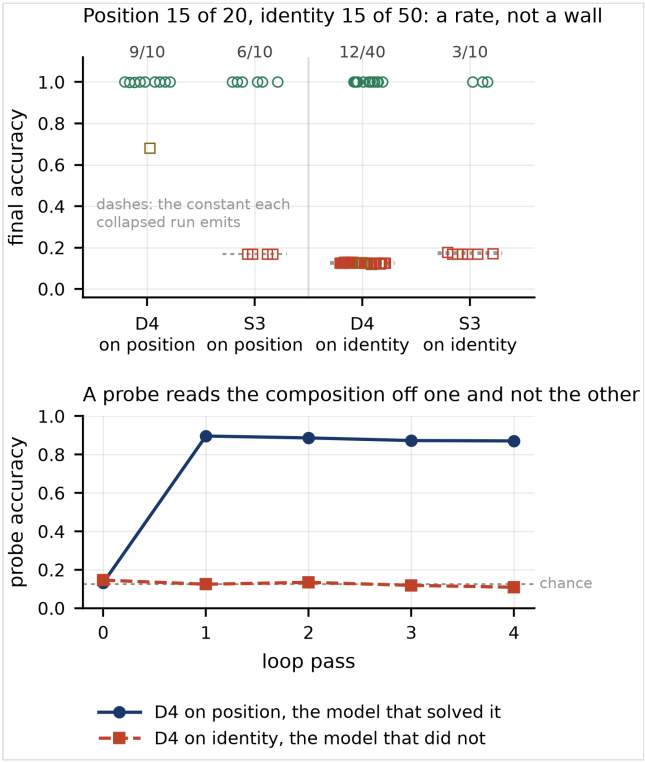


Figure 2. Top, final accuracy by cell of the 2x2, one point per seed. Bottom, a probe's readout of the composite across passes, in a position run that solved the task and an appearance run that did not. Rates are at learning rate 3e-4, weight decay 0.05 and one glyph per patch; the figure writes the appearance condition as identity, its name in the code.

The asymmetry is not about loops. At matched compute the looped model reached 0.9966 and the feedforward baseline 0.9998, and the gate that required looping to win by 0.02 failed. On the D4 appearance task the looped, matched-compute and matched-parameter models escape in 10, 7 and 10 of 30 seeds, with every pairwise Fisher p at or above 0.567.

Two tests, registered before their runs, asked whether the asymmetry survives a change that should not matter. It was not shown to survive either.

change	**prediction**	**outcome**
six learning-rate and weight-decay settings, five seeds a side each	position escapes more often, pooled over settings	**failed**: 10 of 30 against 7 of 30, stratified p = 0.4008
the same, any single setting	appearance never escapes significantly more often	held
glyphs drawn at twice the size, so each spans 2 by 2 patches	position escapes more often	**failed**: 10 of 10 against 7 of 10, p = 0.211

At weight decay 0.5 one run in thirty escaped on either substrate, so half the grid carried little information. That cost the test power, and its registered verdict stands. At the original setting the grid's own five seeds a side went 4 of 5 against 2 of 5. As registered, the asymmetry is shown only at learning rate 3e-4, weight decay 0.05 and one glyph per patch; at no setting did appearance escape significantly more often.

5.3 A training deadlock with a rescue that knows its group

In a failing run the composite is absent, not undecodable. A trained probe reads the composite off the states of the model that solved the appearance task at 0.896, and reads chance off a model that did not, on the same states, at the same sites, with the same code. A null probe is evidence that the information is absent only if the same probe finds it when it is present. **Figure 3** puts the lens row, the probe row and that positive control on one panel for that reason.

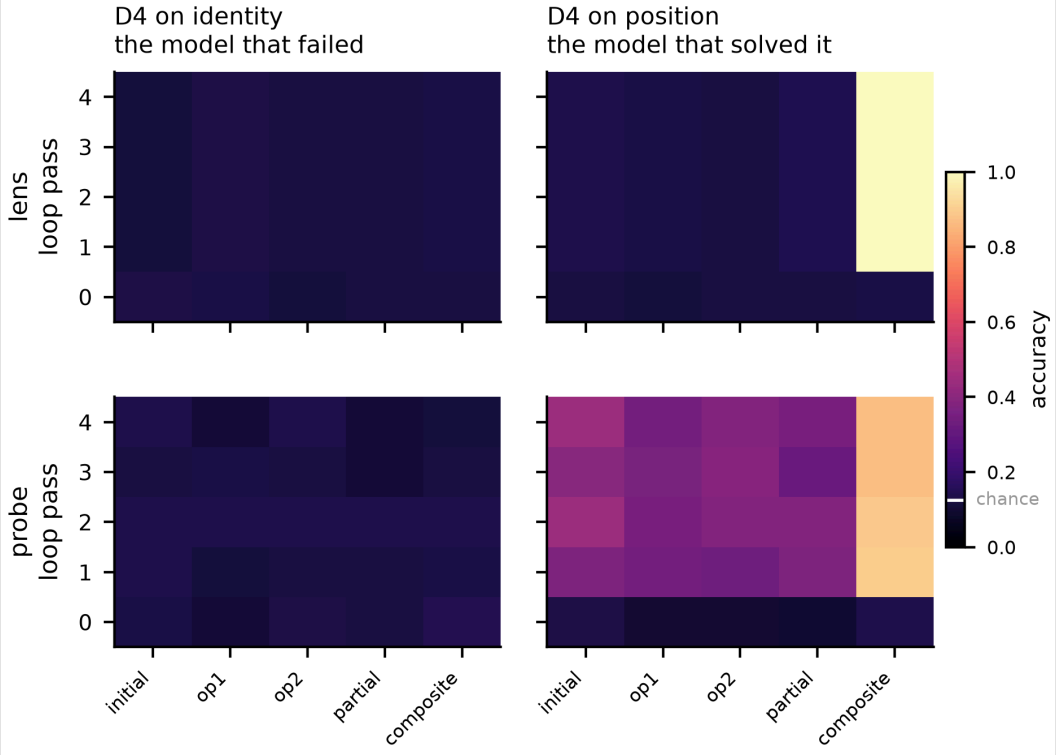


Figure 3. Absent, not undecodable. The same probe at the same sites reads the composite in the solved model and nothing in the failing one; the coda lens row shows the same split.

A donor breaks the deadlock. A depth 2 appearance task that a cold start escapes in 10 of 30 seeds is solved in every one of ten seeds started from a depth 1 donor of itself, each seed from its own donor, and every one of them left chance within about 2000 to 3000 steps.

donor, a depth 1 model of	group	substrate	escaped	Fisher p against the wrong-group donor
the same task	D4	appearance	10 of 10	below 0.0005
the same group on the other substrate	D4	position	9 of 10	0.001

donor, a depth 1 model of	group	substrate	escaped	Fisher p against the wrong-group donor
another group	S3	appearance	1 of 10	
no donor, cold start			10 of 30	0.233

The control that gives this meaning is the wrong-group donor, matched on architecture, prior training and task family and differing only in group. It rescues no more often than a cold start, so the rescue is not generic warm starting. What transfers is the group, not the substrate: the two donors that rescue disagree about substrate, and the one that does not agrees with the task about substrate and differs in group. So, at the setting where the asymmetry holds, how often a task is learned from cold depends on the substrate, and whether a foothold transfers depends on the group.

Both halves held twice on fresh seeds registered in advance.

test	same group, same substrate	same group, other substrate	different group	verdict
first set, five seeds an arm	5 of 5	5 of 5	1 of 5	both held, each p = 0.048
second set, ten seeds an arm	10 of 10	9 of 10	3 of 10	both held, Holm-adjusted p = 0.006 and 0.020

Each test compares one rescuing arm with the wrong-group arm on seeds that did not exist when the test was written. The first set held at the narrowest margin that could pass at five seeds a side; the second, with twice the seeds and a Holm correction across the two tests, held with room to spare. In the second set three wrong-group recipients escaped, two of them within about 2000 steps, a donor's timing rather than a cold start's, and one other-substrate recipient never left chance; both are counted.

If composing appearance were out of the model's representational reach, a donor could not fix it, and that argument does not depend on the rates above.

5.4 A true partial answer that traps training

The staircase explanation failed. Only D4 acting on position climbs a staircase: in 9 of its 10 sealed seeds the training loss sits near ln 4, the level of knowing the answer up to four candidates, for at least 10000 steps before the solve. D4 has three homomorphisms onto the two-element group, and in the position rendering one of them, rotation plus reflection mod 2, is visible as the tile's aspect ratio: with the reflection about the main diagonal, the glyph stands tall exactly when that sum is even. We registered the hypothesis that a visible quotient gives the model partial credit to climb from. Appendix C reports its tests; the one registered as decisive gave the appearance rendering exactly such a quotient.

E2a adds to the appearance rendering a fixed anchor pixel and a cue pixel present exactly when the aspect-ratio character is odd, so the cue composes: the three cue bits of a problem carry one bit about its label. E2b is identical except that its pixel marks something that does not compose.

	composing cue, E2a	matched non-composing pixel, E2b
first set, twenty seeds each: escaped	1 of 20	13 of 20
first set: runs that left chance and climbed to ln 4	14 of 15	0 of 14
second set, ten seeds each: escaped	3 of 10	4 of 10

The trap held as a registered prediction. In the first set, eleven of the fifteen E2a runs that left chance finished between 0.2455 and 0.2495, which is what a model scores if it reads the cue's one bit and nothing else, and one solved. We then registered that at least half of a fresh set's E2a runs that left chance would finish between 0.24 and 0.26, and five of eight did.

In the first set the composing cue escaped far less often than the non-composing pixel. A registered test that this reversal would recur on fresh seeds failed: E2a 3 of 10 against E2b 4 of 10. So the claim here is about where runs that take the cue stop, not that the cue lowers the rate of escape.

This is not ordinary shortcut learning. A shortcut in the usual sense is a feature correlated with the label in the training data and wrong outside it. This cue is a correct quotient of the label everywhere, so a model that uses it is right about part of the answer. The setting has an exact algebraic description of which partial answers exist, and the runs stop at the one that is visible. Our own registered hypothesis expected a visible quotient like this one to help (Appendix C); it gave training a place to stop instead.

5.5 Salience, measured before it was correlated, tracks learnability

How far apart a rendering draws the group's elements was defined before it was computed: the mean, over all pairs of distinct elements, of the Euclidean distance between their rendered tiles in pixels scaled to 0 to 1. The same distance after a freshly initialised model's own patch embedding, averaged over 20 initialisations, was registered as a secondary. The prediction was a Spearman correlation with escape rate of at least 0.6 across seven renderings, with the composing cue left out because Section 5.4 already shows salience does not account for it.

rendering	distance between tiles, pixels	after the patch embedding	escaped
D4 position	2.2455	0.8804	9 of 10
D4 appearance	1.4104	0.5515	12 of 40
D4 moving a pentomino whose box is square	2.3948	0.9391	16 of 20
D4 position tiles assigned to other elements	2.2455	0.8804	14 of 20
D4 appearance with a non-composing pixel	1.6937	0.6602	13 of 20

rendering	distance between tiles, pixels	after the patch embedding	escaped
S3 position	2.2787	0.8875	6 of 10
S3 appearance	1.2868	0.5005	3 of 10

The correlation is 0.673, and the prediction held. D4 position and the rendering that reassigns its tiles draw the same eight tiles, so their distances are equal and the two share a rank, as registered. After the patch embedding the correlation is also 0.673, reported and not tested. Renderings whose elements look further apart are learned more often. With seven points this is not significant: an exact permutation test over all 5040 orderings gives a two-sided p of 0.107. The registration fixed a threshold, not a test, so the prediction held as registered, and we read it as suggestive: salience, measured this way, tracks learnability across these renderings. It was measured at one tile scale, so it does not explain why the asymmetry weakened when glyphs were spread over several patches.

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6. Discussion

The failure is about optimisation, not representation. A donor of the right group fixes it in every seed, a wrong-group donor does no better than a cold start, both held on two sets of fresh seeds, and the probe finds nothing in a failing model that a decoder is merely unable to read. That combination is hard to explain by capacity and easy to explain by bootstrapping: there is no gradient toward the composite until something in the model already computes part of it. A donor supplies that part, and it has to be the right group's part; which substrate it learned the group on does not matter.

Partial credit can be a trap. Handing the model a visible, correct piece of the answer did not lift it toward the whole. Runs that found the piece stopped on it, at the score it earns. The two observations fit together: a donor supplies an internal partial computation that the full task extends, while the cue supplies an external one that the task can be fit around without extending. We have not tested that reading, and it is the experiment we would most want next.

Salience tracks which pictures are learnable, and the substrate effect is smaller than it first looked. The group was learned far more often from position than from colour at the setting we started with, and the renderings whose elements are further apart in pixels are the ones learned more often. But the gap between position and colour did not survive a change of optimisation settings or of tile scale as registered, so we do not claim a general substrate law. What the picture changes may be how far apart it draws the elements to a given tokenizer, rather than whether they are positions or colours.

Looping did not help here, and the failures we study are shared by feedforward models of matched compute and matched parameters. Appendix B reports what the looped models do compute: passes track composition depth and not scene breadth, and the published halting criteria can be scored against an oracle the benchmark provides.

7. Limitations

- The substrate asymmetry is setting-specific. It was not shown to survive a change of learning rate and weight decay or glyphs that span several patches (Section 5.2). Everything else in the paper is at the original setting unless stated.
- The results come from one task family, one renderer and one scale, at width 384 on synthetic images. Whether any result holds at scale or on natural images is untested.
- The salience result is seven points and one measure. It is a rank correlation that cleared a registered threshold with a permutation p of 0.107, not a significant result and not a manipulation of salience, and it was measured at one tile scale.
- The trap claim is about where runs stop, not how often they escape. The composing cue's lower escape rate in the first set did not replicate.
- Held-out confirmation of the asymmetry is "not contradicted", never "confirmed". Three of the four cells have held-out seeds and none contradicts its sealed rate; the S3 position cell has none.
- Some registered tests carry no correction for multiple tests. Each was tested as registered. Under a Holm correction within each registration, the first donor-specificity set ($p = 0.048$ each) and one staircase contrast ($p = 0.018$) would not hold; the second donor-specificity set was registered with Holm and held under it.
- Some comparisons cross experiments that evaluate at different intervals, so they compare distributions and not matched seeds.
- The wrong-group donor result is a rate at ten seeds. It says a wrong-group donor does not help, not whether it harms. Three of ten fresh wrong-group recipients escaped, and why a wrong-group donor ever helps is unexplained.
- The probe evidence is from a failing run. What the appearance runs that do escape extract, and when, is unmeasured.
- Escape rates are rates at the training budget. A shorter budget would misclassify late escapers; one run escaped after step 210000.
- We do not claim looping beats matched compute. That gate failed.

Code and data

Every figure is produced by a script that reads a stored metrics file and computes nothing. Every count in the text comes from a tested generator over the stored runs, and every registered verdict from a tested module that was committed before the runs it reads had finished. The preregistration, every later

registration, a ledger of each registration's commit and date, the stored metrics behind every figure and table, and the figure scripts are released with this paper, at https://github.com/K144U/What-Loops-See.

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Appendix A. Provenance

Each claim, the entry in the released decision log that set up its runs, whether its test was registered before those runs existed, the module that evaluates it, and the stored output it is read from.

claim	decision log entry	test registered before its runs	module	stored output
the 2x2 of Section 5.2	preregistration, `experiment-substrate.md`	yes	`escape_rates`	the runs themselves
held-out cells of the 2x2	D-057	yes	`escape_rates --confirm`	`runs/_why_timing_20260929/confirm_register.txt`
asymmetry across optimisation settings	D-066	yes	`revision_verdicts`	`runs/_revision_verdicts_20261022/`
asymmetry at twice the tile size	D-067	yes	`revision_verdicts`	`runs/_revision_verdicts_20261021/`
looped and feedforward arms	D-032, D-037	no, a comparison of arms	`escape_rates --contrast`	`runs/_curriculum_contrasts_20261006/ffwd_arms.txt`

claim	decision log entry	test registered before its runs	module	stored output
donor arms at ten seeds	D-040	no, confirmed by the next two rows	`escape_timing`, `escape_rates --contrast`	`runs/_why_timing_20260929/timing.txt`, `runs/_curriculum_contrasts_20261006/contrasts.txt`
donor specificity, first fresh set	D-061	yes	`curriculum_confirm`	`runs/_curriculum_confirm_20261009/d061_registered.txt`
donor specificity, second fresh set	D-069	yes	`revision_verdicts`	`runs/_revision_verdicts_20261021/`
the cue renderings, first set	D-062	yes	`why_e1e2`	`runs/_why_e1e2_20261009/`
the trap and the reversal, fresh set	D-068	yes	`revision_verdicts`	`runs/_revision_verdicts_20261021/`
salience and learnability	D-070, D-076	yes	`tile_distinguishability`	`runs/_tile_distinguishability_20261022_d076/`
early stop for added runs	D-065	yes, a protocol	the trainer	`runs/_earlystop_check_20261011/`
every figure, and the numbers of Appendix B	D-074, D-076	not a test; Appendix B is exploratory	`scripts/build_figures.sh`	`runs/_figures_20261022_d076/`

Appendix B. What the looped models compute

These three results were registered under the preregistration; by its own Section 0, the first and third are exploratory. A fourth, on loop-count jitter and wrong attractors, is reported in concurrent work by two of the authors, in submission, and is omitted here.

Passes track depth, not breadth. In ten seeds of family B and ten of family C, the minimum number of passes needed to reach 0.90 rises with composition depth and does not move with scene breadth. Its median is 1, 1, 2, 2, 2 and 3 at depths 1 to 6 in family B. Across breadths 12 to 16 in the same runs, pooled over depth, it stays at 2, and across breadths 4 to 8 in family C it stays at 1. Eight of the 300 family B cells

never reach 0.90 within 32 passes, three at depth 5 and five at depth 6, and are left out of the medians. **Figure 4** plots both axes on a shared scale.

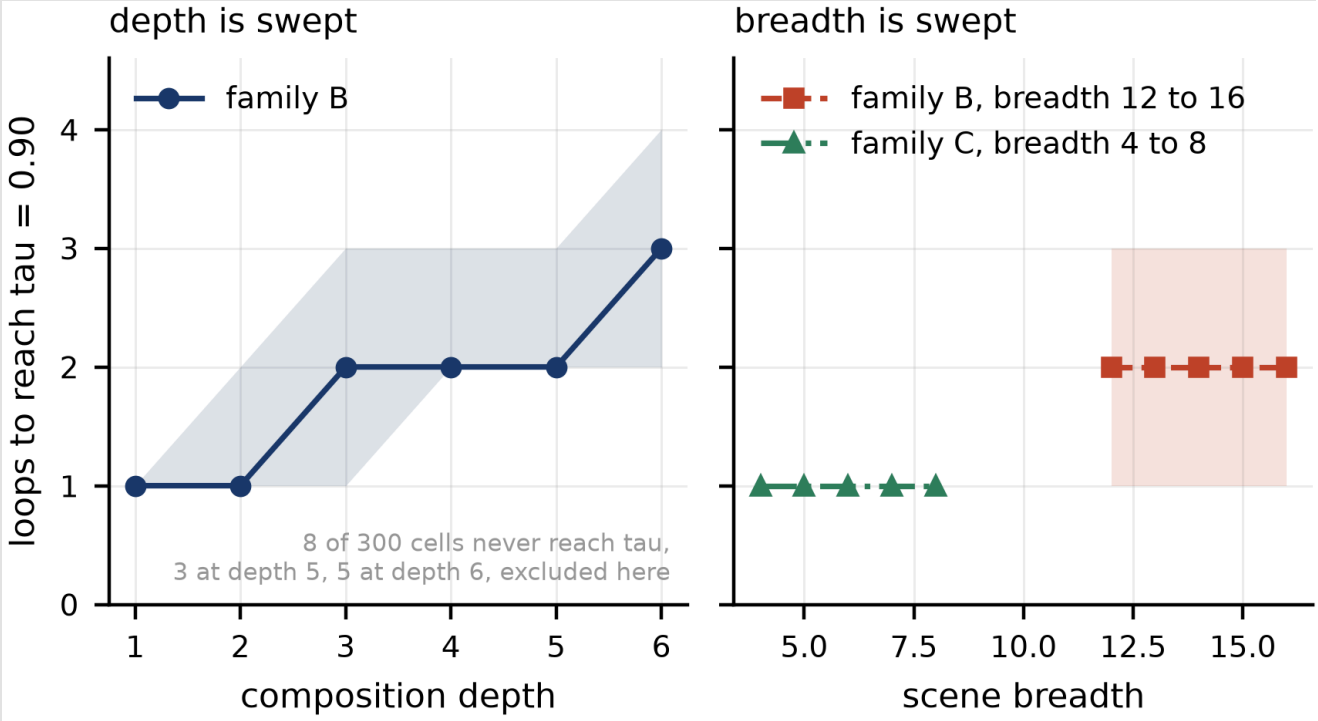


Figure 4. Minimum passes needed against composition depth and against scene breadth, on a shared scale: the median over cells, with the interquartile range shaded.

The benchmark knows how many passes each item needs, so halting criteria can be scored against that rather than against each other. Three of four criteria fire exactly at the model's own requirement, with a median signed error of 0.00. The second-order exit rule of Pappone et al. (2025) fires two loops late, at +2.00 on both families, with a sign test p of 0.0020 against a Holm threshold of 0.00357 across the fourteen tests of that family. Predictive entropy has a median error of zero and a slope of +0.24 against the requirement, firing late on deep items and early on shallow ones so that the errors cancel. **Figure 5** shows every criterion's firing point; its rows are a fixed-pass policy and an adaptive one and are not to be read across. The registered hypothesis, that the criteria are biased against the oracle and the bias grows with depth, needed one criterion with both its error and its slope rejected after Holm, and none has both. The second-order rule's error is rejected, but its slope against depth, at p = 0.0412, is not: four other tests of the family already have larger p values, so no outcome of the rest could let Holm reach it. The two criteria whose slopes are rejected, Kullback-Leibler and entropy, have a median error of exactly zero. And the hypothesis is not falsified, because an error of +2.00 is not zero. It is neither supported nor falsified, and that verdict is determinate.

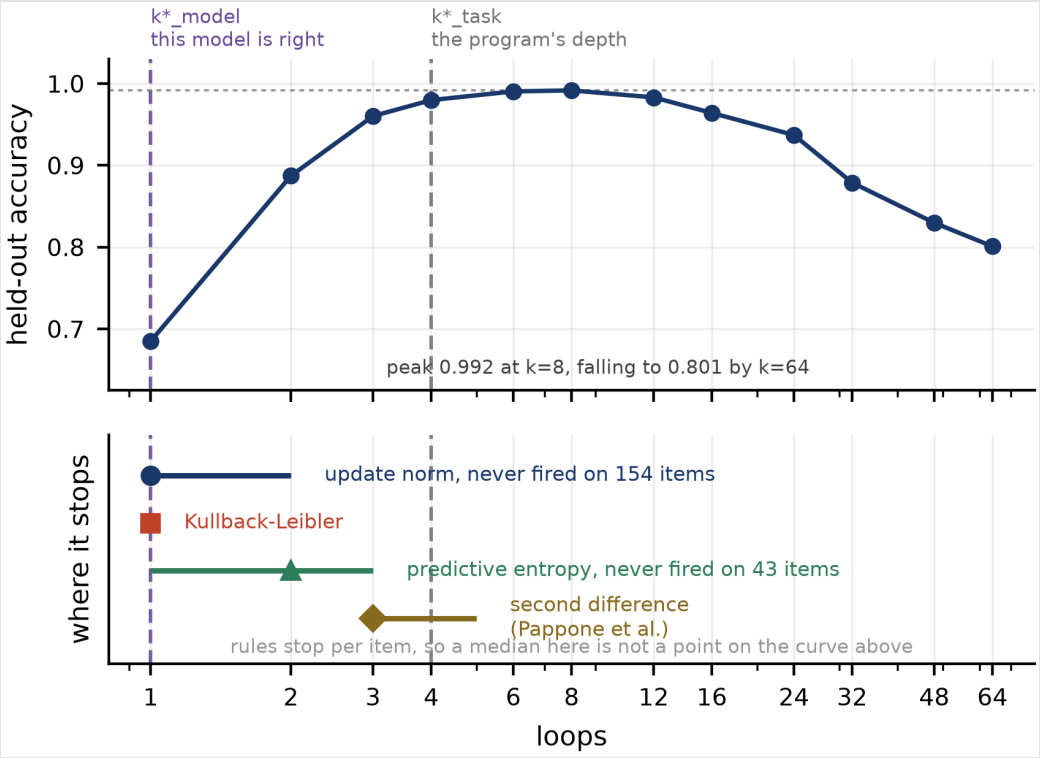


Figure 5. Accuracy against passes with both oracles and every halting criterion's firing point. The rows are a fixed-pass policy and an adaptive one and must not be compared across.

In the patching analysis, on normalised entropy, over 20 depth cells and 10 breadth cells, the loop contrast is null at -0.1021 (confidence interval -0.3062 to +0.1110) and the spatial contrast is significant at -0.1287 (-0.1870 to -0.0754) and opposite to the registered prediction. Neither survives Holm. The null loop contrast pools two depth families that sit on either side of the breadth family, family A's cells mostly below it and family B's mostly above, which **Figure 6** shows for one run of each.

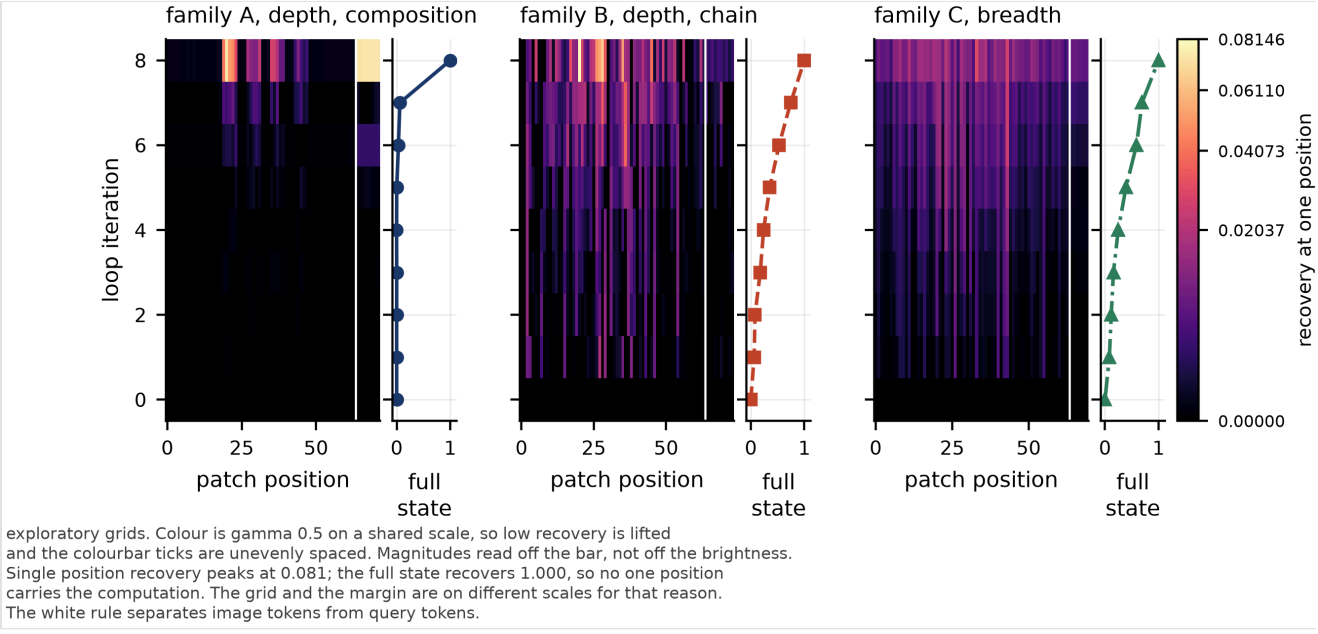


Figure 6. Patching recovery over loop iteration and patch position, one panel per family, with the recovery from patching the full state beside each.

Using the coda lens of Lu et al. (2025), on a depth 6 task the answer rises from the floor to 0.999 while the deepest intermediate never passes 0.201, which replicates their finding in a different modality. **Figure 7** shows the trajectories.

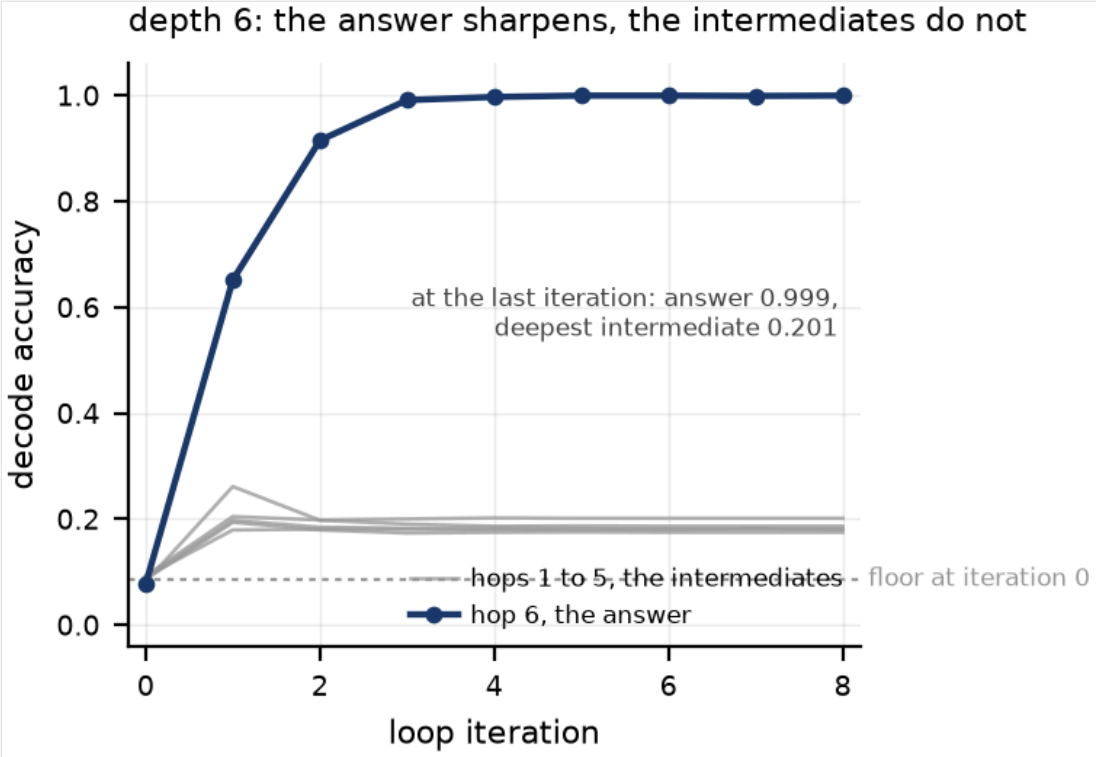


Figure 7. Coda-lens accuracy across passes on a depth 6 task, for the answer and for each intermediate composite.

Appendix C. Registered tests of why the picture matters

Two hypotheses were drawn post hoc from training curves; every test of them was registered before its runs. *H-V, a visible quotient:* composition is learned reliably when some quotient of the group is visible, because partial credit on it builds a pathway that keeps receiving gradient. *H-L, a lottery and a lock:* without such a staircase, escape happens only in an early window that then closes.

prediction	verdict
no held-out appearance seed leaves chance after step 40000	**failed**: one left at step 46350 and solved
at least 4 of 5 held-out D4 position seeds show the ln 4 plateau	held, 4 of 5
no held-out appearance seed sits on a quotient level for 10000 steps	held, none in 10
every D4 position seed with a one-character stage has it on the aspect-ratio character	**failed**: one learned the reflection first
no D4 appearance seed shows a one-character stage	**failed**: two held rotation mod 2

H-L's lock is refuted. H-V's description replicated, but its claim that the appearance rendering offers no accessible foothold did not.

The causal test used four new renderings of D4, twenty seeds each, each changing only what the tiles look like: E1a moves an F-pentomino whose bounding box is square in every orientation, so the aspect-ratio character shows nothing; E1b keeps the position tiles but assigns each to a different element; E2a and E2b are the cue renderings of Section 5.4.

	prediction	**verdict**	**numbers**
V1	E1a escapes less than D4 position	**failed**	16 of 20 against 9 of 10, p = 0.640
V2	E1a climbs less than D4 position	held	7 of 17 against 9 of 10, p = 0.018
V3	E1b escapes less than D4 position	**failed**	14 of 20 against 9 of 10, p = 0.372
V4	E1b climbs less than D4 position	**failed**	12 of 18 against 9 of 10, p = 0.364
V5	E2a, the composing cue, escapes more than E2b	**failed**	E2a 1 of 20, E2b 13 of 20, p below 0.0005
V6	E2a climbs more than E2b	held	E2a 14 of 15, E2b 0 of 14, p below 0.0005

H-V is falsified by the rule fixed in advance: if V5 failed, a visible composing quotient did not make D4 appearance learnable. Among the four accounts registered for this test, the escape pattern matched salience, which Section 5.5 then measured directly.